

Exp.No-6 Time series analysis of website traffic

PROGRAM:

```
import pandas as pd

import matplotlib.pyplot as plt

dataset= {

    "Date": ["2026-01-01", "2026-01-02", "2026-01-03", "2026-01-04", "2026-01-05"],

    "Visitors": [120, 150, 180, 160, 200]

}

df = pd.DataFrame(dataset)

df.to_csv("website.csv", index=False)

print("website.csv created successfully")


data = pd.read_csv("website.csv")


data["Date"] = pd.to_datetime(data["Date"])

data = data.sort_values("Date")


print("Total Visitors:", data["Visitors"].sum())

print("Average Visitors:", data["Visitors"].mean())

print("Maximum Visitors:", data["Visitors"].max())

print("Minimum Visitors:", data["Visitors"].min())


plt.plot(data["Date"], data["Visitors"])

plt.xlabel("Date")

plt.ylabel("Visitors")

plt.title("Website Traffic")

plt.xticks(rotation=45)
```

```
plt.show()
```

OUTPUT:

website.csv created successfully

Total Visitors: 810

Average Visitors: 162.0

Maximum Visitors: 200

Minimum Visitors: 120

